

Intan Fadhilah

Bachelor of Agricultural Engineering | Software, Data, IoT

Bekasi, Indonesia | fadhilahintan003@gmail.com | + 62 81291883156 | miraphira.github.io

PERSONAL

Agricultural Engineering graduate from Universitas Padjadjaran with hands-on experience in software development, data processing, IoT, and automation through academic, research, and personal projects. Experienced in Python, SQL, backend development, database management, API integration, machine learning, and embedded systems. Developed software applications, restructured databases, integrated APIs, built IoT monitoring systems, and implemented automation and control systems. Interested in applying engineering and technology to develop practical solutions and continuously expanding skills across software, data, and smart agricultural technologies.

SKILLS

- **Programming:** Python, SQL, C++, PHP
- **Web & Backend:** FastAPI, HTML, CSS, Jinja2, CRUD Application Development, REST API
- **Database:** MySQL, SQLite, SQL Server, SQL Queries, Database Design & Restructuring
- **Data & Machine Learning:** Data Processing, TensorFlow, Exploratory Data Analysis (EDA)
- **IoT & Embedded Systems:** ESP32, ESP8266, Arduino, Sensor Integration, Real-Time Monitoring
- **Automation & Control:** Automation Systems, ON-OFF Control, PID & Fuzzy Control
- **Tools:** Git, GitHub, Linux, VSCode, MySQL Workbench, Arduino IDE, Figma, Canva, Thinger.io, Cloudflare
- **Problem Solving:** Analytical Thinking, System Analysis, Problem Solving, Project Coordination
- **Languages:** Indonesian (Native), English (Intermediate), Japanese (JLPT N4)

PROJECTS

MODIFICATION OF THE RICE DECISION SUPPORT SYSTEM (DSS) APPLICATION SERVICE BASED ON OPEN SOURCE SOFTWARE FOR PREDICTING NUTRIENT SUFFICIENCY AND DISEASE CONDITIONS Agricultural Engineering Research Project

Sumedang, Indonesia
July 2025 – July 2026

- Redeveloped the DSS Padi application service using an open-source platform to improve system efficiency, maintainability, and integration capabilities.
- Developed Python scripts for database restructuring, data validation, and SQL query optimization.
- Integrated the WhatsApp API to deliver recommendation results to users.
- Developed supporting services for the integration of machine learning applications into the DSS.

PET HEALTH TRACKER Personal Web Project

Sumedang, Indonesia
July 2026 – Present

- Developed a web-based pet health monitoring application using Python, FastAPI, Jinja2, and SQLite.
- Implemented CRUD operations, user authentication, database management, and health-data visualization.
- Designed the application to record and monitor structured health data through an interactive dashboard.

EZMONEY – MACHINE LEARNING CAPSTONE PROJECT

Bangkit Academy – Google Indonesia

Sumedang, Indonesia

November – December 2024

- Developed a personal finance application incorporating machine learning features for savings recommendations and spending anomaly detection.
- Assisted in implementing LSTM (Long Short-Term Memory) model for financial behavior analysis.
- Coordinated team activities throughout the development of a machine learning-based personal finance application, including task coordination, progress monitoring, and communication among team members.

REALTIME MONITORING OF LABORATORY WASTEWATER pH

PT Pupuk Kalimantan Timur Project

Bontang, Indonesia

October – December 2023

- Developed a real-time monitoring system for recording laboratory wastewater pH levels.
- Calibrated pH measuring instruments before use to ensure accurate measurements.
- Implemented continuous monitoring to support water quality assessment and safe discharge management in an industrial environment.
- Worked with sensor-based data acquisition and real-time monitoring technologies.

TEMPERATURE AND HUMIDITY CONTROL SYSTEM

Academic Practical Project

Sumedang, Indonesia

February – April 2024

- Developed a temperature and humidity monitoring and control system using DHT11 and ESP32.
- Implemented threshold-based ON-OFF control to activate or deactivate actuators based on environmental conditions.
- Evaluated system responses under different environmental conditions..

EDUCATION

PADJADJARAN UNIVERSITY

Bachelor of Agricultural Engineer | GPA 3.44

Sumedang, Indonesia

2020-2026

BANGKIT ACADEMY – GOOGLE INDONESIA

Specialized Machine Learning

Jakarta, Indonesia

2024

LICENSES AND CERTIFICATIONS

DICODING INDONESIA

Learning Data Analyst with Python

Credential ID: J LX14LENJX72

Jakarta, Indonesia

Issued October, 2024

Expires October, 2027

Data analysis training fundamental concepts, descriptive statistics, data wrangling, exploratory data analysis (EDA), data visualization, and dashboard development using Streamlit.

GOOGLE

Crash Course on Python

Credential ID: R2P3TPX7860M

Jakarta, Indonesia

Issued September, 2024

Python fundamentals, data structures, scripting, and automation.

GOOGLE

Using Python to Interact with the Operating System

Credential ID: Z4GFPJ0XSYD

Jakarta, Indonesia

Issued September, 2024

Python scripting, file/process manipulation, regular expressions, and Bash.

WORKING EXPERIENCE

PT. PUPUK KALIMANTAN TIMUR

Calibration Laboratory Analyst Intern

Bontang, Indonesia

October 2023 – December 2023

Assisted in calibrating multiple laboratory measuring instruments and regularly recorded calibration data to support measurement accuracy and compliance with standards.

PADJADJARAN UNIVERSITY

Engineering Mathematics Teaching Assistant

Sumedang, Indonesia

September – December 2022

Assisted students in understanding calculus, linear algebra, and differential equations. Supported assignments, examinations, and academic consultations.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

STUDENT EXECUTIVE BOARD OF FTIP UNPAD

Staff of Research and Data Bureau

Sumedang, Indonesia

January – December 2022

Staff of Social and Environment Department

January – December 2021

Staff of Environment Department

October – December 2020

GRAND OPENING PROTEIN FTIP

Deputy Head of Publication, Design, and Documentation Division

Sumedang, Indonesia

August – October 2021

As the Deputy Head of the Publication, Design, and Documentation Division, I supported publication, design, and documentation activities. I coordinated projects, collaborated with team members to create content, and assisted the division head in planning and implementing communication initiatives.

METAMORPH PPMNH

General Secretary

Sumedang, Indonesia

2021 & 2022

As the General Secretary for the New Student Welcome Event, I managed administrative matters across all divisions, including progress records, document preparation, and administrative requirements. I coordinated with division members to collect and organize information and ensure that administrative needs were properly documented and completed.

FTIP MENGABDI

Deputy Field Coordinator

Sumedang, Indonesia

April – October 2022

As the Deputy Field Coordinator, I help plan and execute outreach programs that serve the community. My tasks include organizing logistics, coordinating volunteer efforts, collaborating with local partners, overseeing on-site activities, and ensuring projects meet community needs. I also served as a moderator during community activities, facilitating communication between speakers and participants.

SEKOLAH TANI INDONESIA

Community Volunteer

Bandung, Indonesia

December 2021 – December 2022

Supported community agriculture programs in Cibeunying, Bandung by coordinating between experts and local participants, assisting training on horticulture planting techniques and organic fertilizer production, and helping organize monthly sessions with ~20 participants.